

step01_01_data_quality

October 2, 2026

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[2]: import pandas as pd
import os
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[3]: # List of CSV files in the Train delay and prediction dataset
csv_files = [
    "data_sets/train_details.csv",
    "data_sets/station_full_names.csv",
    "data_sets/combined_schedule.csv",
    "data_sets/combined_delay.csv"
]
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[9]: def describe_datasets_complete(files):
    for file in files:
        if not os.path.exists(file):
            print(f"Dataset '{file}' not found in the current directory.\n")
            continue

        # Load the dataset
        df = pd.read_csv(file, low_memory=False)

        # Calculate total number of tuples (rows) for the entire dataset
        total_tuples = len(df)

        print(f"=== Dataset: {file} ===")
        print(f"Total number of tuples: {total_tuples}\n")

        for column in df.columns:
            # Number of non-empty tuples
            non_empty_count = df[column].notna().sum()

            # Number of unique values
            unique_count = df[column].nunique(dropna=True)

            # List of unique values and the count of each
            value_counts = df[column].value_counts(dropna=True)

            # Truncating the display to top 15 items to handle the ~30 million
            ↪ rows in combined_delay.csv safely[cite: 1]
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        if len(value_counts) > 15:
            display_counts = [f"'{val}': {count}" for val, count in
↪value_counts.head(15).items()]
            display_counts.append("... (truncated)")
        else:
            display_counts = [f"'{val}': {count}" for val, count in
↪value_counts.items()]

        print(f"Attribute: {column}")
        print(f" - Number of non-empty tuples: {non_empty_count}")
        print(f" - Number of missing values:
↪{total_tuples-non_empty_count}")
        print(f" - Number of unique values: {unique_count}")
        print(f" - Unique values and their counts (Top 15): [{', ' .
↪join(display_counts)}]\n")

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[10]: describe_datasets_complete(csv_files)
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=== Dataset: data_sets/train_details.csv ===
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Total number of tuples: 8992
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Attribute: train_no
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- Number of non-empty tuples: 8992
- Number of missing values: 0
- Number of unique values: 8720
- Unique values and their counts (Top 15): ['6526': 2, '5311': 2, '5312': 2,
'56138': 2, '69208': 2, '5509': 2, '5510': 2, '56136': 2, '4849': 2, '4850': 2,
'4851': 2, '4852': 2, '69207': 2, '6269': 2, '4879': 2, ... (truncated)]

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Attribute: train_name
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- Number of non-empty tuples: 8992
- Number of missing values: 0
- Number of unique values: 6455
- Unique values and their counts (Top 15): ['VANDE BHARAT EXP': 172, 'AMRIT
BHARAT EXP': 53, 'INTERCITY EXP': 41, 'RAJYA RANI EXP': 17, 'SHATABDI EXP': 13,
'MAE AMAN SHUTTLE': 11, 'AMAN MAE SHUTTLE': 11, 'PNBE GAYA MEMU': 10, 'JSME BDME
MEMU': 10, 'VANDE BHARAT EX': 10, 'GZB NDLS EMU': 10, 'GAYA PNBE MEMU': 10,
'JANSADHARAN EXP': 10, 'BWN ASN MEMU': 10, 'ASN BWN MEMU': 9, ... (truncated)]

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Attribute: type_code
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- Number of non-empty tuples: 8992
- Number of missing values: 0
- Number of unique values: 8
- Unique values and their counts (Top 15): ['PASS-TRAINS': 4538, 'EXP-TRAINS':
2255, 'SF-TRAINS': 1382, 'PRM-TRAINS': 448, 'T18-TRAINS': 198, 'SHT-TRAINS': 64,
'GRB-TRAINS': 54, 'RAJ-TRAINS': 53]

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=== Dataset: data_sets/station_full_names.csv ===

Total number of tuples: 8963

Attribute: station_name

- Number of non-empty tuples: 8963
- Number of missing values: 0
- Number of unique values: 8963
- Unique values and their counts (Top 15): ['ZW': 1, 'AA': 1, 'AABH': 1, 'YS': 1, 'YPR': 1, 'YPD': 1, 'YP': 1, 'YNRK': 1, 'YNK': 1, 'YNG': 1, 'YLM': 1, 'YLK': 1, 'YLG': 1, 'YLBA': 1, 'YL': 1, ... (truncated)]

Attribute: station_full_name

- Number of non-empty tuples: 8963
- Number of missing values: 0
- Number of unique values: 8940
- Unique values and their counts (Top 15): ['KAMALAPUR': 3, 'KARJARA': 2, 'JAGDISHPUR': 2, 'GOVINDGARH': 2, 'HALLIKHED K': 2, 'BELA': 2, 'GANGADHARA': 2, 'DHAULPUR': 2, 'HIWARKHED': 2, 'KOLAGHAT': 2, 'BAREILLY': 2, 'AMMUGUDA': 2, 'DADAR': 2, 'MOHANA': 2, 'NAGARI': 2, ... (truncated)]

Attribute: station_zone

- Number of non-empty tuples: 8963
- Number of missing values: 0
- Number of unique values: 19
- Unique values and their counts (Top 15): ['NR': 1047, 'ECR': 781, 'SCR': 758, 'SR': 735, 'WR': 705, 'ER': 587, 'NWR': 586, 'NFR': 558, 'NER': 501, 'CR': 468, 'NCR': 425, 'SWR': 400, 'SER': 373, 'ECOR': 340, 'SECR': 311, ... (truncated)]

Attribute: station_address

- Number of non-empty tuples: 8869
- Number of missing values: 94
- Number of unique values: 6996
- Unique values and their counts (Top 15): ['Uttar Pradesh, India': 134, 'Madhya Pradesh, India': 68, 'Bihar, India': 43, 'Rajasthan, India': 38, 'Gujarat, India': 32, 'India': 30, 'Maharashtra, India': 28, 'Pali, Rajasthan, India': 18, 'Rae Bareli, Uttar Pradesh, India': 17, 'Bharatpur, Rajasthan, India': 17, 'Andhra Pradesh, India': 17, 'Chhindwara, Madhya Pradesh, India': 15, 'Nagpur, Maharashtra, India': 15, 'Hardoi, Uttar Pradesh, India': 14, 'Jalgaon, Maharashtra, India': 14, ... (truncated)]

=== Dataset: data_sets/combined_schedule.csv ===

Total number of tuples: 172112

Attribute: station_no

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 119
- Unique values and their counts (Top 15): ['1': 8673, '2': 8673, '3': 8542,

'4': 8417, '5': 8285, '6': 8082, '7': 7853, '8': 7556, '9': 7242, '10': 6890, '11': 6543, '12': 6187, '13': 5859, '14': 5530, '15': 5154, ... (truncated)]

Attribute: station_name

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 8629
- Unique values and their counts (Top 15): ['BZA': 469, 'BRC': 438, 'CNB': 436, 'ST': 405, 'ET': 398, 'BSL': 391, 'DDU': 359, 'NDLS': 343, 'LKO': 339, 'KYN': 323, 'ADI': 312, 'VGLJ': 311, 'PNBE': 310, 'UMB': 303, 'BBS': 294, ... (truncated)]

Attribute: distance_from_origin

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 3241
- Unique values and their counts (Top 15): ['0': 8674, '7': 870, '6': 864, '23': 846, '33': 835, '9': 829, '39': 811, '20': 805, '10': 803, '12': 796, '21': 790, '26': 789, '28': 782, '19': 781, '11': 779, ... (truncated)]

Attribute: arrival_day

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 4
- Unique values and their counts (Top 15): ['1': 123978, '2': 41849, '3': 5898, '4': 387]

Attribute: arrival_time

- Number of non-empty tuples: 163439
- Number of missing values: 8673
- Number of unique values: 1440
- Unique values and their counts (Top 15): ['09:00': 303, '09:30': 302, '08:30': 293, '09:10': 291, '07:50': 288, '19:00': 284, '19:30': 284, '07:30': 283, '10:10': 280, '20:00': 280, '18:00': 279, '10:00': 273, '08:20': 271, '18:30': 268, '06:50': 266, ... (truncated)]

Attribute: departure_day

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 4
- Unique values and their counts (Top 15): ['1': 123910, '2': 41902, '3': 5913, '4': 387]

Attribute: departure_time

- Number of non-empty tuples: 163439
- Number of missing values: 8673
- Number of unique values: 1440
- Unique values and their counts (Top 15): ['08:00': 454, '08:20': 427,

'07:00': 427, '06:00': 426, '07:30': 424, '18:00': 423, '17:00': 415, '19:00': 413, '16:30': 412, '08:40': 399, '06:50': 398, '06:30': 398, '18:50': 397, '18:40': 396, '07:20': 394, ... (truncated)]

Attribute: train_no

- Number of non-empty tuples: 172112
- Number of missing values: 0
- Number of unique values: 8673
- Unique values and their counts (Top 15): ['13031': 119, '53041': 118, '13032': 117, '53042': 114, '13007': 112, '13049': 111, '13050': 106, '19024': 97, '19023': 94, '58002': 94, '58001': 94, '18237': 92, '13120': 90, '15910': 90, '13119': 89, ... (truncated)]

=== Dataset: data_sets/combined_delay.csv ===

Total number of tuples: 38428703

Attribute: date

- Number of non-empty tuples: 38428703
- Number of missing values: 0
- Number of unique values: 365
- Unique values and their counts (Top 15): ['2025-12-26': 107614, '2025-12-19': 107556, '2025-12-12': 107552, '2026-01-30': 107518, '2025-11-21': 107469, '2025-12-29': 107461, '2026-01-23': 107414, '2026-01-16': 107384, '2025-11-28': 107379, '2025-12-22': 107329, '2025-12-15': 107328, '2025-12-05': 107317, '2025-11-14': 107308, '2026-02-06': 107284, '2025-12-08': 107252, ... (truncated)]

Attribute: station_no

- Number of non-empty tuples: 38428703
- Number of missing values: 0
- Number of unique values: 121
- Unique values and their counts (Top 15): ['1': 1895758, '2': 1895758, '3': 1875376, '4': 1846160, '5': 1823631, '6': 1779462, '7': 1737908, '8': 1662456, '9': 1598984, '10': 1527709, '11': 1451640, '12': 1377087, '13': 1311977, '14': 1231539, '15': 1146932, ... (truncated)]

Attribute: station_name

- Number of non-empty tuples: 38428703
- Number of missing values: 0
- Number of unique values: 8227
- Unique values and their counts (Top 15): ['NDLS': 79739, 'CNB': 78218, 'BZA': 71156, 'DLI': 63828, 'PNBE': 63747, 'GZB': 62339, 'UMB': 61116, 'BRC': 59879, 'ST': 58095, 'ASN': 54966, 'HWH': 54919, 'SBC': 53807, 'DDU': 52740, 'ET': 52335, 'BWN': 52164, ... (truncated)]

Attribute: delay

- Number of non-empty tuples: 36554432
- Number of missing values: 1874271

- Number of unique values: 4940
- Unique values and their counts (Top 15): ['0.0': 4898984, '1.0': 1244411, '3.0': 1050327, '2.0': 992081, '4.0': 921420, '5.0': 919153, '6.0': 845702, '7.0': 816307, '8.0': 787537, '10.0': 782852, '9.0': 758192, '11.0': 704436, '12.0': 682438, '13.0': 658190, '15.0': 642568, ... (truncated)]

Attribute: train_no

- Number of non-empty tuples: 38428703
- Number of missing values: 0
- Number of unique values: 7033
- Unique values and their counts (Top 15): ['13031': 44165, '13032': 43800, '18237': 33945, '15910': 33945, '15909': 33945, '13152': 32850, '13352': 32850, '19038': 32850, '13151': 32485, '19037': 32485, '13351': 31755, '18238': 31390, '13553': 31025, '13554': 31025, '11058': 31025, ... (truncated)]

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